Rakshit Rameshbabu 24BYB1117
Database Systems Lab-6,7

Sub-queries and Views

A. SINGLE-ROW SUBQUERY (First 2)

1. Employees whose salary exceeds the most recently joined employee

```
SQL> SELECT emp_name, salary
  2  FROM EMPLOYEE_24BYB1117
  3  WHERE salary > (
  4      SELECT salary
  5      FROM EMPLOYEE_24BYB1117
  6      WHERE join_date = (
  7          SELECT MAX(join_date)
  8          FROM EMPLOYEE_24BYB1117
  9      )
 10  );

EMP_NAME                          SALARY
------------------------------ ----------
Anand                            45000.75
Bhavya                            38000.4
Charan                            52000.6
Divya                               60000
Ezhil                             40000.2
```

2. Employees whose age is higher than the youngest employee

```
SQL> SELECT emp_name, age
  2  FROM EMPLOYEE_24BYB1117
  3  WHERE age > (
  4      SELECT MIN(age)
  5      FROM EMPLOYEE_24BYB1117
  6  );

EMP_NAME                                AGE
----------------------------- ----------
Anand                                    28
Bhavya                                   32
Charan                                   26
Divya                                    35
Ezhil                                    29
```

B. MULTI-ROW SUBQUERY (First 2)

1. Students who scored more marks than students of semester 4

```
SQL> SELECT DISTINCT student_id, marks
  2  FROM ENROLLMENT_24BYB1117
  3  WHERE marks > ALL (
  4      SELECT marks
  5      FROM ENROLLMENT_24BYB1117
  6      WHERE semester = 4
  7  );

STUDENT_ID      MARKS
---------- ----------
       103         75
       101         85
       103         90
       102         68
       101         78
       105         88
       105         82
       102         72

8 rows selected.
```

2. Employees whose salary is lower than salaries of IT department employees

```
SQL> SELECT emp_name, salary
  2  FROM EMPLOYEE_24BYB1117
  3  WHERE salary < ALL (
  4      SELECT salary
  5      FROM EMPLOYEE_24BYB1117
  6      WHERE dept = 'IT'
  7  );

no rows selected
```

C. MULTIPLE-COLUMN SUBQUERY (First 2)

1. Students enrolled in the same course and semester as student Arun

```
SQL> SELECT *
  2  FROM ENROLLMENT_24BYB1117
  3  WHERE (course_id, semester) IN (
  4      SELECT e.course_id, e.semester
  5      FROM ENROLLMENT_24BYB1117 e
  6      JOIN STUDENT_24BYB1117 s
  7      ON e.student_id = s.student_id
  8      WHERE s.student_name = 'Arun'
  9  );

 ENROLL_ID STUDENT_ID  COURSE_ID      MARKS   SEMESTER
---------- ---------- ---------- ---------- ----------
         1        101        201         78          3
         2        101        204         85          3
         3        102        201         68          3
         8        105        201         88          3
```

2. Employees sharing same department and salary as another employee

```
SQL> SELECT emp_name, dept, salary
  2  FROM EMPLOYEE_24BYB1117
  3  WHERE (dept, salary) IN (
  4      SELECT dept, salary
  5      FROM EMPLOYEE_24BYB1117
  6      GROUP BY dept, salary
  7      HAVING COUNT(*) > 1
  8  );

no rows selected
```

D. CORRELATED SUBQUERY (First 2)

1. Employees who earn more than all others in their department

```
SQL> SELECT emp_name, dept, salary
  2  FROM EMPLOYEE_24BYB1117 e
  3  WHERE salary > ALL (
  4      SELECT salary
  5      FROM EMPLOYEE_24BYB1117
  6      WHERE dept = e.dept
  7        AND emp_id <> e.emp_id
  8  );

EMP_NAME                       DEPT                    SALARY
------------------------------ -------------------- ----------
Charan                         IT                     52000.6
Divya                          FINANCE                  60000
Ezhil                          HR                     40000.2
```

2.Students who secured highest marks in each course

```
SQL> SELECT student_id, course_id, marks
  2  FROM ENROLLMENT_24BYB1117 e
  3  WHERE marks = (
  4      SELECT MAX(marks)
  5      FROM ENROLLMENT_24BYB1117
  6      WHERE course_id = e.course_id
  7  );

STUDENT_ID  COURSE_ID      MARKS
---------- ---------- ----------
       103        204         90
       103        202         75
       104        205         60
       105        201         88
       105        203         82
```

A. SIMPLE VIEW QUESTIONS (First 5)

1. View to display customer name and city only

```
SQL> CREATE VIEW customer_city_view_24BYB1117 AS
  2  SELECT customer_name, city
  3  FROM CUSTOMER_24BYB1117;

View created.
```

2. View showing account number, account type, and balance

```
SQL> CREATE VIEW account_basic_view_24BYB1117 AS
  2  SELECT account_no, account_type, balance
  3  FROM ACCOUNT_24BYB1117;

View created.
```

3. View to display customers belonging to Chennai city

```
SQL> CREATE VIEW chennai_cust_v_24BYB1117 AS
  2  SELECT *
  3  FROM CUSTOMER_24BYB1117
  4  WHERE city = 'Chennai';

View created.
```

4. View showing savings account holders only

```
SQL> CREATE VIEW sav_acc_v_24BYB1117 AS
  2  SELECT *
  3  FROM ACCOUNT_24BYB1117
  4  WHERE account_type = 'Savings';

View created.
```

5. View to display customers with balance above ₹1,00,000

```
SQL> CREATE VIEW high_bal_v_24BYB1117 AS
  2  SELECT *
  3  FROM ACCOUNT_24BYB1117
  4  WHERE balance > 100000;

View created.
```

B. COMPLEX VIEW QUESTIONS (First 5)

1. View showing customer name, account number, and balance

```
SQL> CREATE VIEW cust_acc_v_24BYB1117 AS
  2  SELECT c.customer_name, a.account_no, a.balance
  3  FROM CUSTOMER_24BYB1117 c
  4  JOIN ACCOUNT_24BYB1117 a
  5  ON c.customer_id = a.customer_id;

View created.
```

2. View showing customer name, branch name, and loan amount

```
SQL> CREATE VIEW cust_loan_v_24BYB1117 AS
  2  SELECT c.customer_name, l.branch_name, l.loan_amount
  3  FROM CUSTOMER_24BYB1117 c
  4  JOIN LOAN_24BYB1117 l
  5  ON c.customer_id = l.customer_id;

View created.
```

3. View to display branch-wise total account balance

```
SQL> CREATE VIEW branch_tot_v_24BYB1117 AS
  2  SELECT branch_name, SUM(balance) AS total_balance
  3  FROM ACCOUNT_24BYB1117
  4  GROUP BY branch_name;

View created.
```

4. View showing customer-wise total loan amount

```
SQL> CREATE VIEW cust_totloan_v_24BYB1117 AS
  2  SELECT customer_id, SUM(loan_amount) AS total_loan
  3  FROM LOAN_24BYB1117
  4  GROUP BY customer_id;

View created.
```

5. View to display customers who have both an account and a loan

```
SQL> CREATE VIEW cust_acc_loan_v_24BYB1117 AS
  2  SELECT DISTINCT c.customer_name
  3  FROM CUSTOMER_24BYB1117 c
  4  JOIN ACCOUNT_24BYB1117 a
  5  ON c.customer_id = a.customer_id
  6  JOIN LOAN_24BYB1117 l
  7  ON c.customer_id = l.customer_id;

View created.
```

Queries in Green colour

```
SQL> SELECT emp_name, salary
  2  FROM EMPLOYEE_24BYB1117
  3  WHERE salary > (
  4    SELECT AVG(salary)
  5    FROM EMPLOYEE_24BYB1117
  6  );

EMP_NAME                           SALARY
------------------------------ ----------
Charan                              52000
Divya                               60000

SQL> SELECT student_id, marks
  2  FROM ENROLLMENT_24BYB1117
  3  WHERE marks > ANY (
  4    SELECT marks
  5    FROM ENROLLMENT_24BYB1117
  6    WHERE semester = 4
  7  );

no rows selected

SQL> SELECT student_name
  2  FROM STUDENT_24BYB1117
  3  WHERE student_id IN (
  4    SELECT student_id
  5    FROM ENROLLMENT_24BYB1117
  6    WHERE course_id = 201
  7  );

STUDENT_NAME
------------------------------
Arun
Banu

SQL> SELECT emp_name, dept, salary
  2  FROM EMPLOYEE_24BYB1117 e
  3  WHERE salary = (
  4    SELECT MAX(salary)
  5    FROM EMPLOYEE_24BYB1117
  6    WHERE dept = e.dept
  7  );

EMP_NAME                       DEPT                     SALARY
------------------------------ -------------------- ----------
Bhavya                         HR                        38000
Charan                         IT                        52000
Divya                          FINANCE                   60000
```

```
SQL> CREATE VIEW cust_city_v_24BYB1117 AS
  2  SELECT customer_name, city
  3  FROM CUSTOMER_24BYB1117;

View created.

SQL> CREATE VIEW elec_prod_v_24BYB1117 AS
  2  SELECT product_name, price
  3  FROM PRODUCT_24BYB1117
  4  WHERE category = 'Electronics';

View created.

SQL> CREATE VIEW cust_ord_v_24BYB1117 AS
  2  SELECT c.customer_name, o.order_date, o.total_amount
  3  FROM CUSTOMER_24BYB1117 c
  4  JOIN ORDERS_24BYB1117 o
  5  ON c.customer_id = o.customer_id;

View created.

SQL> CREATE VIEW cat_avg_v_24BYB1117 AS
  2  SELECT category, AVG(price) AS avg_price
  3  FROM PRODUCT_24BYB1117
  4  GROUP BY category;

View created.
```